Zifeng Zhao, PhD

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EDUCATION

Ph.D. **Tongji University** **Sep.2018 – Dec.2023**
Research Focus: Asphalt Materials, Reclaimed Asphalt Pavement, Pavement Performance
Advisor: Prof. Feipeng Xiao

Visiting Ph.D. **Politecnico di Milano** **Nov.2022 – Oct.2023**
Research Focus: Sustainable Asphalt Mixtures, Viscoelastic Characterization, Pavement Recycling
Advisor: Prof. Maurizio Crispino, Prof. Emanuele Toraldo

B.E. **Chang'an University** **Sep.2014 – Jun.2018**

RESEARCH EXPERIENCE

Postdoctoral Research Associate **Jul.2024 till now**
Department of Civil, Environmental, and Geo-Engineering, University of Minnesota, Twin Cities

- Investigate asphalt mixture compaction, low-temperature cracking, and long-term pavement performance through laboratory testing, MnROAD field evaluations, and finite element modeling.
- Develop physics-based and data-driven methods for pavement condition assessment using embedded temperature sensors, wavelet analysis, inverse heat analysis, and machine learning.
- Contribute to externally funded research projects in collaboration with the Minnesota Department of Transportation (MnDOT), addressing asphalt pavement durability, recycled materials, and infrastructure maintenance.
- Collaborate with researchers from Michigan State University, the Federal Highway Administration (FHWA), the National Center for Asphalt Technology (NCAT), and Università Politecnica delle Marche (Italy) on pavement materials, performance evaluation, and computational modeling.

Doctoral Researcher **Sep.2018 – Dec.2023**
Department of Transportation Engineering, Tongji University

- Investigated the aging and viscoelastic behavior of rubberized asphalt and reclaimed asphalt pavement (RAP) mixtures under coupled temperature, moisture, and ultraviolet radiation using accelerated aging, rheological testing, and mechanical characterization.
- Evaluated rutting, cracking, and strain responses of flexible, semi-flexible, and rigid pavements through full-scale accelerated pavement testing (APT), embedded strain sensors, and statistical analysis.

Doctoral Researcher **Nov.2022 – Oct.2023**
Department of Civil Engineering, Politecnico di Milano

- Collaborated on international research into recycled asphalt mixtures, focusing on RAP incorporation, viscoelastic performance, and multi-field aging effects to support sustainable pavement applications.

Laboratory Research and Management Assistant **Sep.2020 – Oct.2022**
Key Laboratory of Aviation Facility Durability and Operational Safety, Tongji University

- Supported applied airfield pavement research involving recycled construction and demolition materials, laboratory performance evaluation, and field implementation for airport pavement construction and rehabilitation.
- Led the construction of an indoor accelerated pavement testing (APT) test section, including experimental design, embedded strain sensor installation, and three-month pavement performance monitoring.

Research Assistant **Sep.2014 – Jun.2018**
School of Highway, Chang'an University

- Participated in undergraduate research and laboratory investigations related to pavement materials and transportation infrastructure engineering.

TEACHING EXPERIENCE

Guest Lecturer/Teaching Assistant, CEGE 4253 – Pavement Engineering and Management **2024 – 2026**

Department of Civil, Environmental, and Geo-Engineering, University of Minnesota, Twin Cities

- Delivered classroom lectures on selected pavement engineering topics to graduate students.
- Developed and revised lecture materials, including PowerPoint presentations, and supported course instruction in collaboration with the course instructor.
- Graded assignments and examinations, maintained student grade records, and held office hours to address student questions on pavement design.

Guest Lecturer/Teaching Assistant, CEGE 3402 – Civil Engineering Materials **2024 – 2025**

Department of Civil, Environmental, and Geo-Engineering, University of Minnesota, Twin Cities

- Delivered classroom lectures on selected pavement engineering topics to undergraduate students, including cement material properties and cement concrete pavement design.
- Assisted with laboratory instruction, experimental demonstrations, and student guidance in materials testing and data interpretation.

Graduate Teaching Assistant, Advanced Pavement Materials **2019 – 2020**

Department of Transportation Engineering, Tongji University

- Assisted with teaching activities and academic instruction in transportation and pavement engineering.
- Contributed to the development of pavement structural design software.

STUDENT RESEARCH MENTORING

Graduate Research Mentor

University of Minnesota, Twin Cities

- Mentored a master's student on aggregate image analysis and characterization for pavement material research.

Graduate Research Mentor

Tongji University

- Mentored master's students in the research of recycling construction and demolition waste in airfield cushion layer.
- Mentored undergraduate students in asphalt materials research, experimental investigation, data analysis, and research activities.

FELLOWSHIPS & SCHOLOARSHIPS

China Scholarship Council (CSC) Fellowship (approx. $17,400) 2022 – 2023

- Competitive fellowship supporting a 12-month visiting doctoral research at Politecnico di Milano, Italy.

Doctoral Academic Scholarship, Tongji University (approx. $6,000) 2018 – 2023

- Tuition-support scholarship awarded throughout doctoral study.

Yan Ke-Fei Scholarship for Doctoral Student, Tongji University ($500) 2023

Scholarship of China Communication First Highway Engineering ($500) 2016

Central-North Scholarship, Chang'an University ($500) 2016

Ningxia Highway & Road Scholarship, Chang'an University ($500) 2015

HONORS & AWARDS

Outstanding Doctoral Dissertation Award, Tongji University 2023

- University-wide recognition awarded to no more than 10% of doctoral dissertations annually.

Outstanding Graduate, Chang'an University 2018

Outstanding Undergraduate Thesis and Design Award, Chang'an University 2018

- University-level recognition; approximately 3% selection quota under documented university criteria.

PUBLICATIONS

Peer-Reviewed Journal Articles

[J.27] **Zhao, Z*.**, Turos, M. & Marasteanu, M. (2026) Low-Temperature Behavior of Asphalt Binders: Analyzing Field and Laboratory Oxidative Ageing Using Multi-Metric Tests, *Journal of Materials in Civil Engineering*, DOI: 10.1061/JMCEE7/MTENG-25337. (in production)

[J.26] **Zhao, Z*.**, Marasteanu, M., Zanko, L., Turos, M. & Peterson, E. (2026) Comparative Evaluation of Testing Methods for Pavement Marking Friction Differential, *Journal of Traffic and Transportation Engineering (English Edition)*. (in production)

[J.25] **Zhao, Z*.**, DeDene, C., Turos, M., Rossow, E., Marasteanu, M. & Dave, E. (2026) Thermal Conductivity of Reclaimed Asphalt Pavement and its Constituents using Finite Element Modeling Approach, *Road Materials and Pavement Design*, 1-21. DOI: 10.1080/14680629.2026.2689544

[J.24] **Zhao, Z*.**, DeDene, C., Turos, M., Marasteanu, M. & Dave, E. (2026) Finite Element Modeling of Particle-Scale Heating of Reclaimed Asphalt Pavement (RAP) in Drum Plants Considering Size, Material, and Internal Temperature Gradients, *Construction and Building Materials in Transportation*, 1-2: 100007. DOI: 10.1016/j.jcbmtr.2026.100007

[J.23] Guo, Y., Xu, L., Chen, X. & **Zhao, Z*.** (2026) A Two-Stage Feature Reduction (FIRRE) Framework for Improving Artificial Neural Network Predictions in Civil Engineering, *Infrastructures*, 11(1), 29. DOI: 10.3390/infrastructures11010029

[J.22] Xu, L., **Zhao, Z.**, Lai, Y., Yuan, Y., Wang, S., Lin, J., Moretti, L. & Loprencipe, G*. (2025) Effect of Bonding Characteristics on Rutting Resistance and Moisture Susceptibility of Rubberized Reclaimed Asphalt Pavement, *Infrastructures*, 10(12), 336. DOI: 10.3390/infrastructures10120336

[J.21] **Zhao, Z.**, Xiao, F*., Toraldo, E., Crispino, M., & Antoniazzi, A. (2025). Radiation-Thermal-Moisture Coupling Effects on Viscoelasticity Properties of Reclaimed Asphalt Rejuvenated by Rubberized Asphalt. *Construction and Building Materials*, 470, 140535. DOI: 10.1016/j.conbuildmat.2025.140535

[J.20] **Zhao, Z.**, Han, L., Bao, K., Zhang, Z. & Xiao, F*. (2025) Investigation of Viscoelasticity Variation of Rubberized Asphalt Rejuvenated Reclaimed Asphalt Pavement Materials under Multi Fields Coupling Effects, *China Journal of Highway and Transport*, 38(3): 224-240. DOI: 10.19721/j.cnki.1001-7372.2025.03.017 (in Chinese)

[J.19] Wu, J., **Zhao, Z.**, Jiang, C., Yang, Y., Sun, Z., Yuan, J., & Xiao, F*. (2024). Recent Development and Application of Natural Fiber in Asphalt Pavement. *Journal of Cleaner Production*, 449, 141832. DOI: 10.1016/j.jclepro.2024.141832

[J.18] He, W., **Zhao, Z.**, Zhang, Z., & Xiao, F*. (2024). Modification and Application of the Arrhenius Equation based on Activated Energy Methods from Various Asphalt Binders. *Journal of Materials in Civil Engineering*, 36(6), 04024123. DOI: 10.1061/JMCEE7.MTENG-17482

[J.17] Wu, J., **Zhao, Z.**, Jiang, C., Cao, Z., Xu, X., & Yuan, J*. (2024). Mechanical and Durability Characteristics of Recycled Brick-Concrete Materials and Their Application in Airport Pavement Cushion. *Journal of Tongji University (Natural Science)*, 52(9): 1374-1383. DOI: 10.11908/j.issn.0253-374x.24123.

[J.16] **Zhao, Z.**, Xiao, F*., Toraldo, E., Crispino, M., & Ketabdari, M. (2023). Effect of Crumb Rubber and Reclaimed Asphalt Pavement on Viscoelastic Property of Asphalt Mixture. *Journal of Cleaner Production*, 428, 139422. DOI: 10.1016/j.jclepro.2023.139422.

[J.15] **Zhao, Z.**, Xu, L., Li, X., Guan, X., & Xiao, F*. (2023). Comparative Analysis of Pavement Performance Characteristics of Flexible, Semi-Flexible and Rigid Pavement based on Accelerated Pavement Tester. *Construction and Building Materials*, 387, 131672. DOI: 10.1016/j.conbuildmat.2023.131672

[J.14] Xu, L., Magar, S., **Zhao, Z.**, Xiang, Q., & Xiao, F*. (2023). Rheological and Anti-Moisture Characteristics of Rubberized Reclaimed Asphalt Pavement with Interfacial Bond Behavior. *Journal of Cleaner Production*, 391, 136172. DOI: 10.1016/j.jclepro.2023.136172

[J.13] Xiao, F*., Xu, L., **Zhao, Z.**, & Hou, X. (2023). Recent Applications and Developments of Reclaimed Asphalt Pavement in China, 2010–2021. *Sustainable Materials and Technologies*, 37, e00697. DOI: 10.1016/j.susmat.2023.e00697

[J.12] He, W., **Zhao, Z.**, Yuan, J., & Xiao, F*. (2023). Recent Development of Ethylene-Vinyl Acetate Modified Asphalt. *Construction and Building Materials*, 363, 129800. DOI: 10.1016/j.conbuildmat.2022.129800

[J.11] Xu, L., Han, Y., Yan, B., **Zhao, Z.**, Li, X., & Xiao, F*. (2023). Investigation on the Stability and Interaction of Asphalt Emulsion with Waterborne Acrylate as Pavement Maintenance Coating. *Progress in Organic Coatings*, 174, 107234. DOI: 10.1016/j.porgcoat.2022.107234

[J.10] **Zhao, Z.**, Xu, L., Guan, X., Li, X., & Xiao, F*. (2022). Rutting and Strain Characteristics of Rubberized Asphalt Pavement based on Accelerated Pavement Tester. *Journal of Cleaner Production*, 376, 134219. DOI: 10.1016/j.jclepro.2022.134219

[J.9] **Zhao, Z.**, Li, J., Xiao, X., Li, Z., & Xiao, F*. (2022). Statistical Analysis on Grading Characteristics between Dense and Stone Matrix Asphalt Mixtures. *Journal of Materials in Civil Engineering*, 34(8), 04022177. DOI: 10.1061/(ASCE)MT.1943-5533.0004322

[J.8] **Zhao, Z.**, Xu, L., Du, Z., & Xiao, F*. (2022). Moisture Resistance of Stone Matrix Asphalt at Lab Simulated High Temperature and Continuous Rainfall Condition. *International Journal of Pavement Engineering*, 24(2): 2096884. DOI: 10.1080/10298436.2022.2096884

[J.7] Xu, L., **Zhao, Z***., Li, J., Li, X., Jiang, C., & Xiao, F*. (2022). Thermal Anti-Cracking Investigation of Asphalt-based Seal Technology for Airport Base Layer in Frozen Ground Region. *Cold Regions Science and Technology*, 204, 103673. DOI: 10.1016/j.coldregions.2022.103673

[J.6] **Zhao, Z.**, Wang, J., Hou, X., Xiang, Q., & Xiao, F*. (2021). Viscosity Prediction of Rubberized Asphalt-Rejuvenated Recycled Asphalt Pavement Binders using Artificial Neural Network Approach. *Journal of Materials in Civil Engineering*, 33(5), 04021071. DOI: 10.1061/(ASCE)MT.1943-5533.0003679

[J.5] Xu, L., **Zhao, Z.**, Li, X., Yuan, J., Zhou, Q., & Xiao, F. (2021). Cracking Investigation on Fog Seal Technology with Waterborne Acrylate and Polyurethane as a Clean Modification Approach. *Journal of Cleaner Production*, 329, 129751. DOI: 10.1016/j.jclepro.2021.129751

[J.4] Hou, X., Hettiarachchi, C., Xiao, F., **Zhao, Z.**, Xiang, Q., & Yong, D. (2021). Blending efficiency improvement and energy investigation of recycled asphalt mixture involved warm mix technology. *Journal of Cleaner Production*, 279, 123732. DOI: 10.1016/j.jclepro.2020.123732

[J.3] **Zhao, Z.**, Xiao, F*., & Amirkhanian, S. (2020). Recent Applications of Waste Solid Materials in Pavement Engineering. *Waste Management*, 108, 78–105. DOI: 10.1016/j.wasman.2020.04.024

[J.2] **Zhao, Z.**, Guan, X., Xiao, F*., Xie, Z., & Zhou, Q. (2020). Applications of Asphalt Concrete Overlay on Portland Cement Concrete Pavement. *Construction and Building Materials*, 264, 120045. DOI: 10.1016/j.conbuildmat.2020.120045

[J.1] Xiang, Q., Hou, X., **Zhao, Z.**, Prozzi, J., & Xiao, F. (2020). Comparative Evaluation of Methods for Removing Residual Mineral Fillers During Bitumen Extraction and Recovery based on FTIR. *Journal of Materials in Civil Engineering*, 32(12), 04020357. DOI: 10.1061/(ASCE)MT.1943-5533.0003452

Peer-Reviewed Conference Proceedings

[C.2] **Zhao, Z***., Khanal, S., Marasteanu, M. & Le, J.-L. (2026). Assessing Fine Aggregate Morphology and Its Link to FAA in Superpave Asphalt Mix Design. In: Pasetto, M., Pasquini, E., Cannone Falchetto, A., Giacomello, G., Baliello, A. (eds) Proceedings of the 2nd RILEM International Symposium on Bituminous Materials (ISBM 2026). ISBM 2026. RILEM Book series, vol 75. Springer, Cham. DOI: 10.1007/978-3-032-30655-5_44

[C.1] **Zhao, Z.**, Li, J., Xiao, X., Li, Z., & Xiao, F*. (2021). Passing rate difference analysis on dense and stone matrix gradations of asphalt mixtures. In *Green and Intelligent Technologies for Sustainable and Smart Asphalt Pavements: Proceedings of the 5th International Symposium on Frontiers of Road and Airport Engineering, pp. 253–259. CRC Press. DOI: 10.1201/9781003251125-41

Technical Reports

[R.4] Le, J.-L., Marasteanu, M., Turos, M., **Zhao, Z.**, & Khanal, S. (2026). Investigation on Mix Design of Recycled Asphalt Pavement (RAP) Materials. Minnesota Department of Transportation, St. Paul, Minnesota. Report No. MN 2026-21.

[R.3] Marasteanu, M., Levin, M., **Zhao, Z.**, Nouhi, S., & Turos, M. (2026). Haul Road and Detour Maintenance. Minnesota Department of Transportation, St. Paul, Minnesota. Report No. MN 2026-06

[R.2] Marasteanu, M., Le, J.-L., **Zhao, Z.**, Liu, Y., Turos, M., Barman, M. & Bhatt, B. (2026). Sawing and Sealing Joints in Bituminous Pavements to Control Cracking. Minnesota Department of Transportation, St. Paul, Minnesota. Report No. MN 2026-23.

[R.1] R. M. dos Santos, K., Marasteanu, M., **Zhao, Z.**, G.C.S. Duarte, Joao., Custis, S. (2025). A Computational Tool for the Reliable Prediction of Pavement Performance based on Temperature Data. Minnesota Department of Transportation, St. Paul, Minnesota. Report No. MN 2025-40.

Books & Book Contributions

[B.2] Xiao, F. (2025). Theory and Applications of Pavement Engineering Materials. China Communications Press. ISBN: 9787114197451. Chapter 5 contributor: "Applications of Solid Waste Materials in Pavement Engineering." (in Chinese).

[B.1] Xiao, F. (2021). Theory and Methods of Asphalt Materials Recycling. Tongji University Press. ISBN: 9787560897554. Chapter 6 contributor: "Characterization of Rubberized Asphalt Mixture." (in Chinese).

Patents

[P.2] Dense-Skeleton Micropore-Sealed Cement-Stabilized Construction Waste Material: Preparation Method and Applications. Granted Chinese Invention Patent. Application No. CN202311158061.6.

[P.1] Airfield Pavement Structure Incorporating High Contents of Construction and Demolition Waste for High-Moisture Silt Subgrades. Granted Chinese Invention Patent. Application No. CN202311158330.9.

CONFERENCE PRESENTATION

Oral Presentation

[C.12] **Zhao, Z**., Khanal, S., Marasteanu, M., & Le, J.-L. (2026). Assessing Fine Aggregate Morphology and its Link to FAA in Superpave Asphalt Mix Design. 2nd RILEM International Symposium on Bituminous Materials (ISBM 2026), Padua, Italy.

[C.11] Zhao, Z., Tianhao, Y., Marasteanu, M., Le, J.-L. & Turos, M. (2026). Understanding the Effect of Asphalt Lift Thickness on Compaction, Density Uniformity, and Ride Quality. CTS Transportation Research Conference, Minneapolis, Minnesota, USA

[C.10] Marasteanu, M., Le, J.-L., **Zhao, Z**. & Turos, M. (2026). Experimental Investigation on Compaction Performance of Asphalt Mixtures with High RAP Content. International Symposium on Pavements for Carbon Neutral and Digitalized Roads (ISPACAD 2026), Granada, Spain.

[C.9] **Zhao, Z**., Khanal, S., Le, J.-L., Marasteanu, M., & Turos, M. (2025). Experimental Investigation on Compaction Properties of Asphalt Mixtures with High RAP Content. CTS Transportation Research Conference, Minneapolis, Minnesota, USA.

[C.8] **Zhao, Z**., Xu, L., Guan, X., Li, X., & Xiao, F. (2022). Rutting and strain characteristics of rubberized asphalt pavement with accelerated pavement testing. Advances in Materials and Pavement Performance Prediction (AM3P), Hong Kong, China.

[C.7] **Zhao, Z**., Li, J., Xiao, X., Li, Z., & Xiao, F. (2021). Passing rate difference analysis on dense and stone matrix gradations of asphalt mixtures. 5th International Symposium on Frontiers of Road and Airport Engineering (IFRAE), Delft, Netherlands.

Poster Presentation

[C.6] **Zhao, Z**., Rossow, E., Yelizarov, N., Feng, X., & Marasteanu, M. (2026). Investigating Tree Shade Effect on Pavement Temperature. 63rd Petersen Asphalt Research Conference, Laramie, WY, USA.

[C.5] **Zhao, Z**., Duarte, J. G. C. S., Marasteanu, M., & dos Santos, K. (2026). Using Long-Term Embedded Temperature Data and Spectral Analysis to Evaluate Performance Trends for Rigid Pavement. 105th Transportation Research Board (TRB) Annual Meeting, Washington, D.C., USA.

[C.4] **Zhao, Z**., Marasteanu, M. & Turos, M. (2026). Using Performance Curves to Evaluate Pavement Damage from Detours: A Case Study on CSAH 5 in Minnesota. 105th Transportation Research Board (TRB) Annual Meeting, Washington, D.C., USA.

[C.3] Marasteanu, M., Zanko, L., **Zhao, Z**., Turos, M. & Peterson, E. (2026). Using Continuous Friction Measurement Technology to Measure Pavement Markings Friction Differential: A MnROAD Case Study. 105th Transportation Research Board (TRB) Annual Meeting, Washington, D.C., USA.

[C.2] **Zhao, Z**., Xiao, F., Toraldo, E., Crispino, M., & Antoniazzi, A. (2025). Radiation-Thermal-Moisture Coupling Effects on Viscoelasticity Properties of Reclaimed Asphalt Rejuvenated by Rubberized Asphalt. 104th Transportation Research Board (TRB) Annual Meeting, Washington, D.C., USA.

[C.1] He, W., **Zhao, Z**., Zhang, Z., Cao, Z., & Xiao, F. (2024). Modification and application of the Arrhenius equation based on activated energy methods from various asphalt binders. 103rd Transportation Research Board (TRB) Annual Meeting, Washington, D.C., USA.

RESEARCH GRANTS & FUNDED PROJECTS

University of Minnesota, Twin Cities, USA

[1] Impact of Asphalt Lift Thickness on Pavement Density and Durability 2024 – Present

- Funding Agency: Minnesota Department of Transportation (MnDOT)
- Role: Research Team Member
- Investigate thick-lift asphalt compaction, density uniformity, mechanical performance, and numerical modeling to support efficient pavement construction.

[2] Tools to improve the durability of asphalt pavements 2025 – Present

- Funding Agency: Minnesota Department of Transportation (MnDOT)
- Role: Research Team Member
- Evaluate compaction performance of Warm Mix Asphalt (WMA) collected from various sources, and characterize the cracking resistance of WMA.

[3] Investigation on Mix Design of Recycled Asphalt Pavement (RAP) Materials 2024 – 2026

- Funding Agency: Minnesota Department of Transportation (MnDOT), Local Road Research Board (LRRB)
- Role: Research Team Member
- Evaluate RAP mixture design, gyratory compaction behavior, and low-temperature cracking resistance to support the use of higher RAP contents.

[4] A Computational Tool for the Reliable Prediction of Pavement Performance Based on Temperature Data 2024 – 2025

- Funding Agency: Minnesota Department of Transportation (MnDOT)
- Role: Research Team Member
- Develop sensor-based and physics-informed analytical methods for long-term pavement condition assessment using embedded temperature measurements.

[5] Haul Road and Detour Maintenance 2024 – 2026

- Funding Agency: Minnesota Department of Transportation (MnDOT)
- Role: Research Team Member
- Evaluate pavement deterioration, structural condition, and maintenance needs associated with construction haul roads and traffic detours.

Tongji University, China

[6] Multi-Field Coupling Evolution Characteristics and Durability of Rubberized Asphalt-Rejuvenated RAP Mixtures in Desert Conditions 2019 – 2021

- Funding Agency: National Natural Science Foundation of China (NSFC), Academy of Scientific Research and Technology (ASRT), Egypt
- Role: Key Research Team Member
- Investigated coupled thermal, moisture, and radiation ageing effect on the rheological and viscoelastic properties of recycled asphalt materials..

[7] Comparative Analysis of Pavement Performance Characteristics of Flexible, Semi-Flexible, and Rigid Pavements Based on Accelerated Pavement Testing 2021 – 2022

- Funding Agency: National Natural Science Foundation of China (NSFC)
- Role: Key Research Team Member
- Evaluated pavement rutting, cracking, and strain responses through accelerated loading tests and embedded sensor measurements.

[8] Secondary Reconstruction of Airfield and Subgrade at Jinan Yaoqiang International Airport 2020 – 2021

- Funding Agency: China Aviation Industry Research Funding
- Role: Key Research Team Member
- Led experimental planning and sensor layout design, including technical drawings and on-site geotechnical investigations for airfield pavement and subgrade reconstruction.
- Coordinated and conducted laboratory testing and material performance evaluation to support pavement rehabilitation design and the application of recycled construction and demolition wastes.

ACADEMIC & PROFESSIONAL SERVICE

Editorial Service

[1] **Guest Editor**, *Infrastructures* (MDPI)

- Special Issue: Pavement Performance and Maintenance: Smart Technologies and Sustainable Practices

[2] Guest Editor, *Frontiers in Materials* (Frontiers)

- Research Topic: "Functional Materials for Sustainable Pavement Engineering: Multi-scale Design and Environmental Integration."

Journal Reviewer

- *International Journal of Pavement Research and Technology* (2022 – Present)
- *Expert Systems with Applications* (2024 – Present)
- *Transportation Engineering* (2024 – Present)
- *Journal of Environmental Management* (2024 – Present)
- *Infrastructures, Applied Sciences, Polymers, Materials* (2025 – Present)
- *Construction and Building Materials* (2025 – Present)
- *Journal of Road Engineering* (2025 – Present)
- *Cleaner Materials* (2026 – Present)
- *Construction and Building Materials in Transportation* (2026 – Present)
- *Journal of Traffic and Transportation Engineering (English Edition)* (2026 – Present)
- *Case Studies in Construction Materials* (2026 – Present)
- *Progress in Engineering Science* (2026 – Present)
- *Automation in Construction* (2026 – Present)
- *Materials and Structures* (2026 – Present)

Conference Organization

- Organizing Committee, International Doctoral Forum on Highway and Airfield Engineering, Tongji University, Shanghai, China (2019, 2021)
- Organizing Committee, Sustainable Pavement and Pavement Recycling Conference, Tongji University, Shanghai, China (2021)

TECHNICAL & RESEARCH SKILLS

Pavement Materials Testing

- Asphalt binder rheology (DSR, BBR)
- Asphalt mixture mechanical testing (UTM, MTS)
- Chemical Characterization (accelerated aging, FTIR, GPC)
- Moisture susceptibility testing (HWT, ITT)
- Low temperature cracking testing (DENT, SCB, DT)

Pavement Performance Evaluation

- Accelerated Pavement Testing (APT)
- Embedded sensor instrumentation and monitoring
- Friction testing

Computational Modeling & Data Analysis

- Finite Element Modeling (FEM)
- Artificial Neural Networks (ANN)
- Statistical Modeling
- Image Processing
- Large Language Modeling (BERTopic)
- LiDAR data processing

Programming & Software

- Python, MATLAB, FreeFEM++, ParaView, SPSS